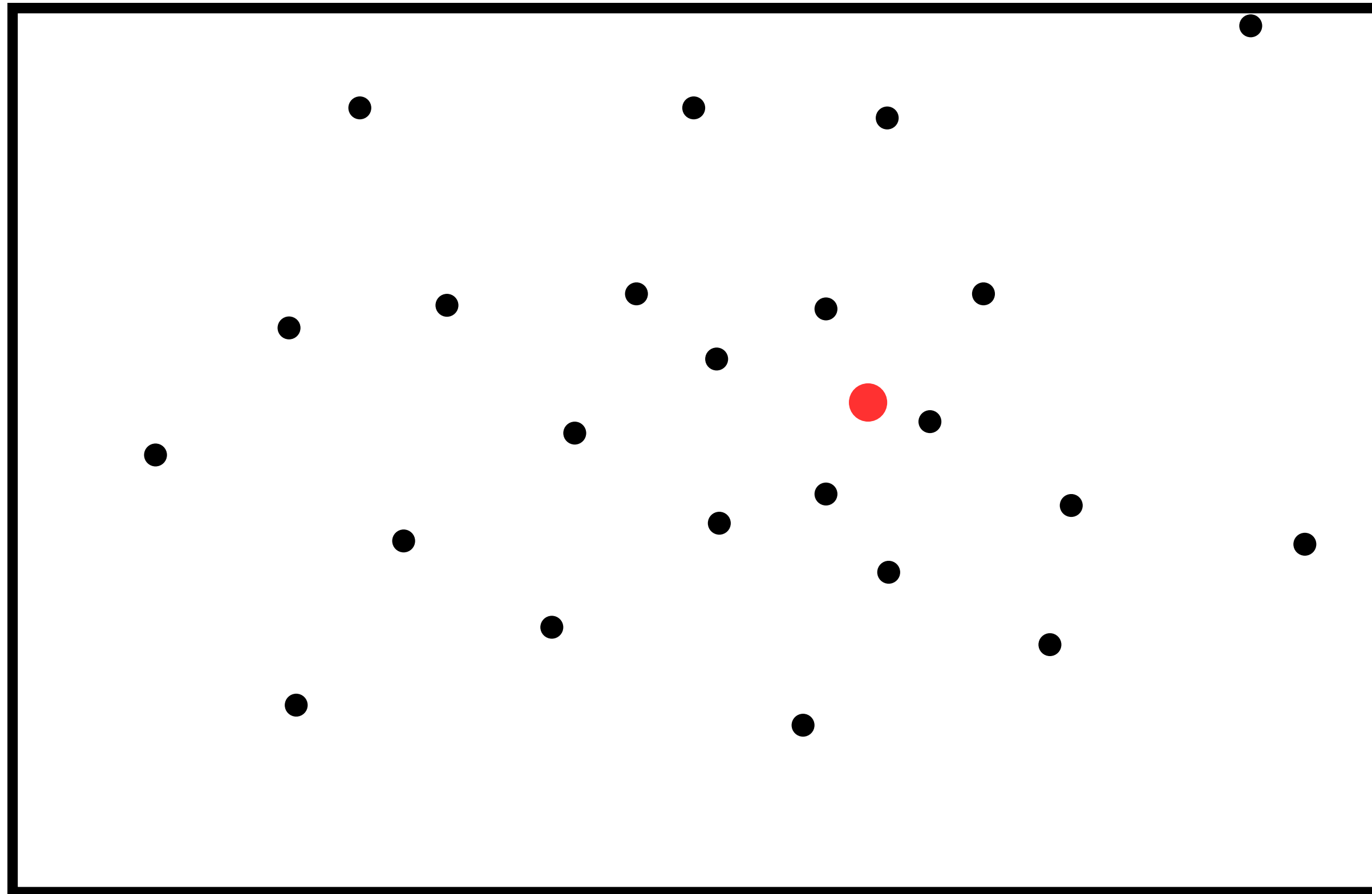




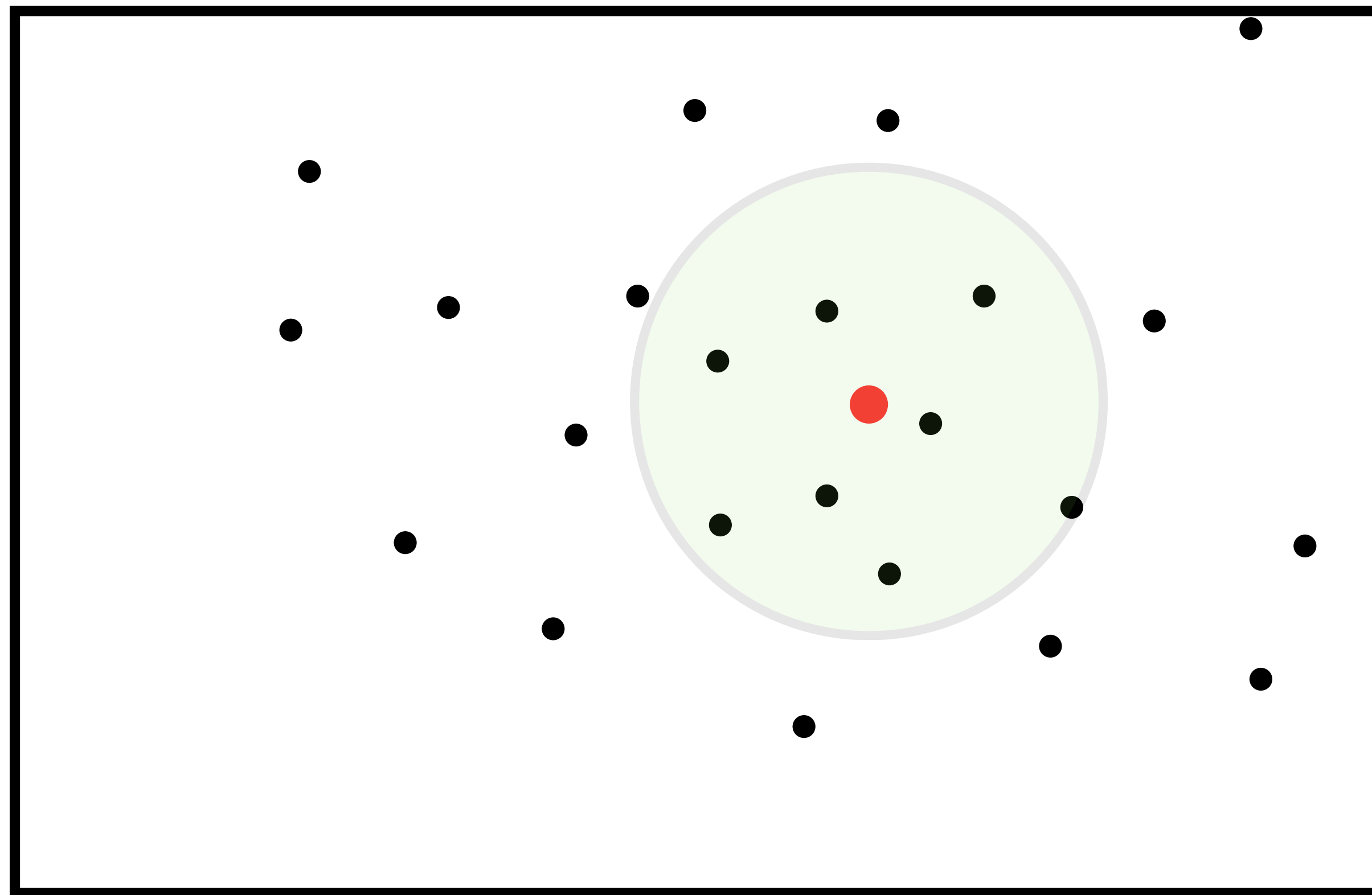
World



You



Restaurants



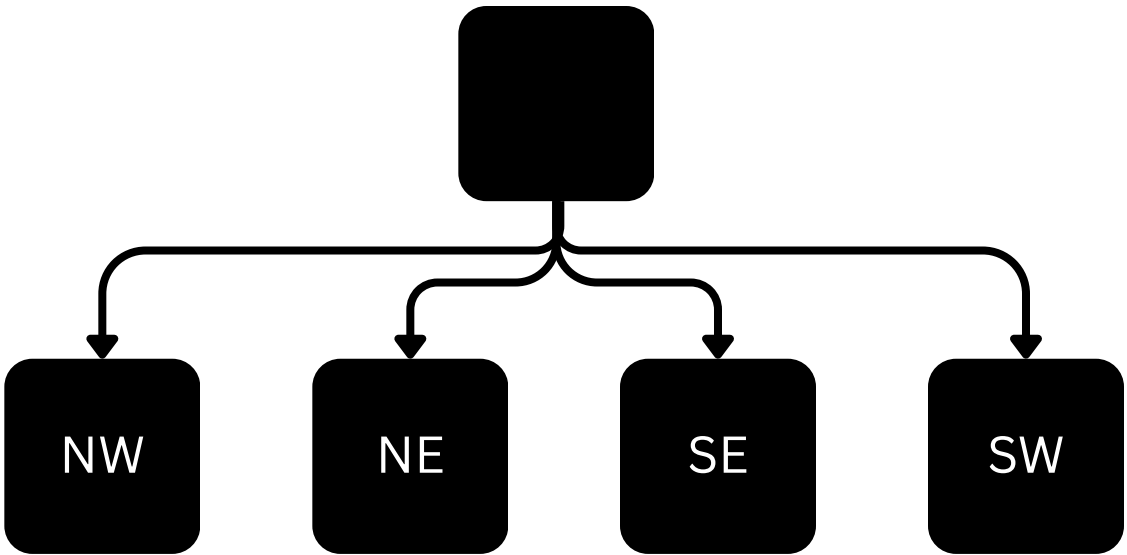
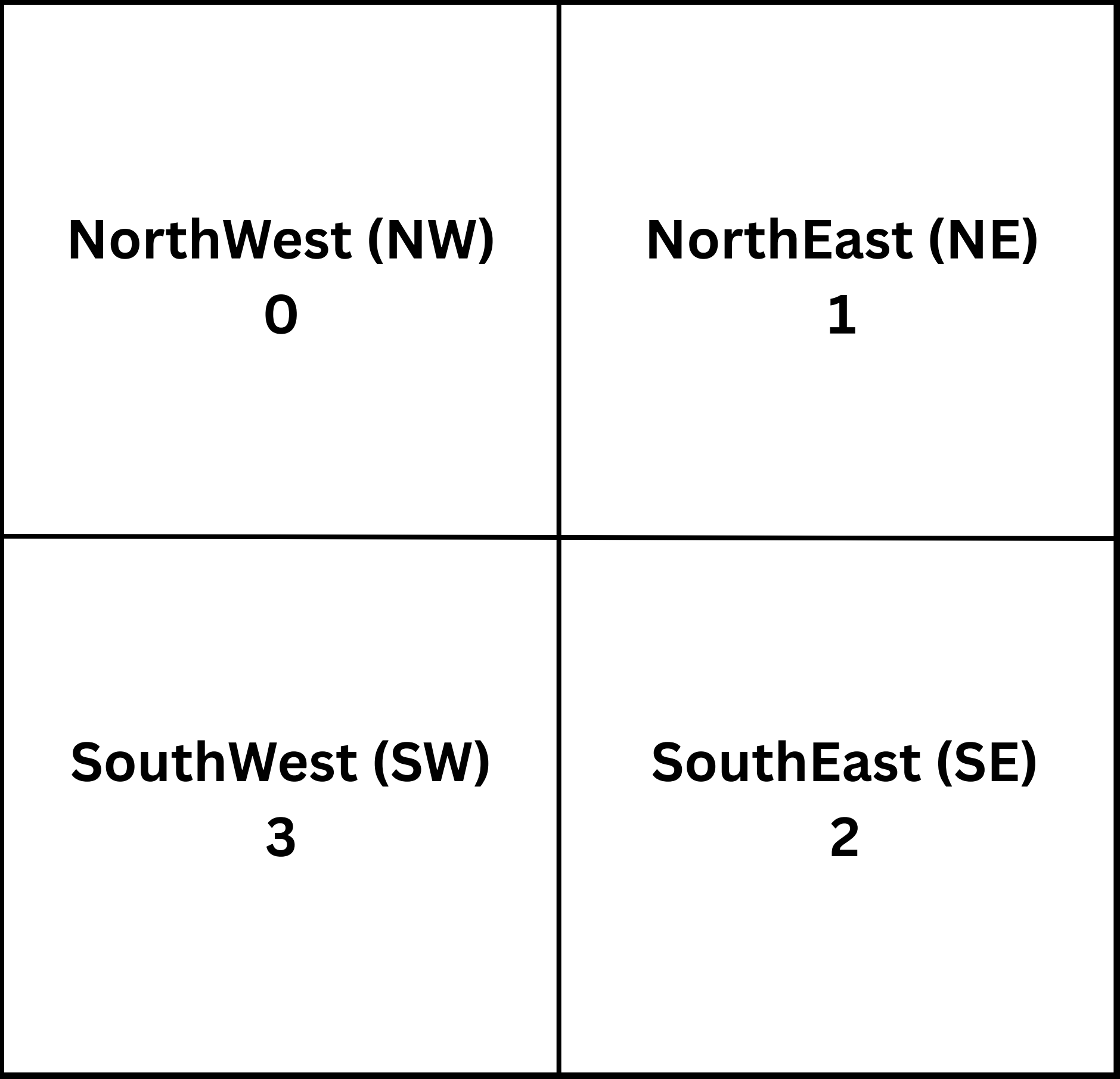
-  World
-  You
-  Restaurants
-  Search area

QuadTree

Contents

1. **QuadTree Definition**
2. **Point Insertion Algorithm**
3. **Range Query Algorithm**
4. **Use Cases**
5. **When not to use**

QuadTree Definition



QuadTree 

NW 

NE 

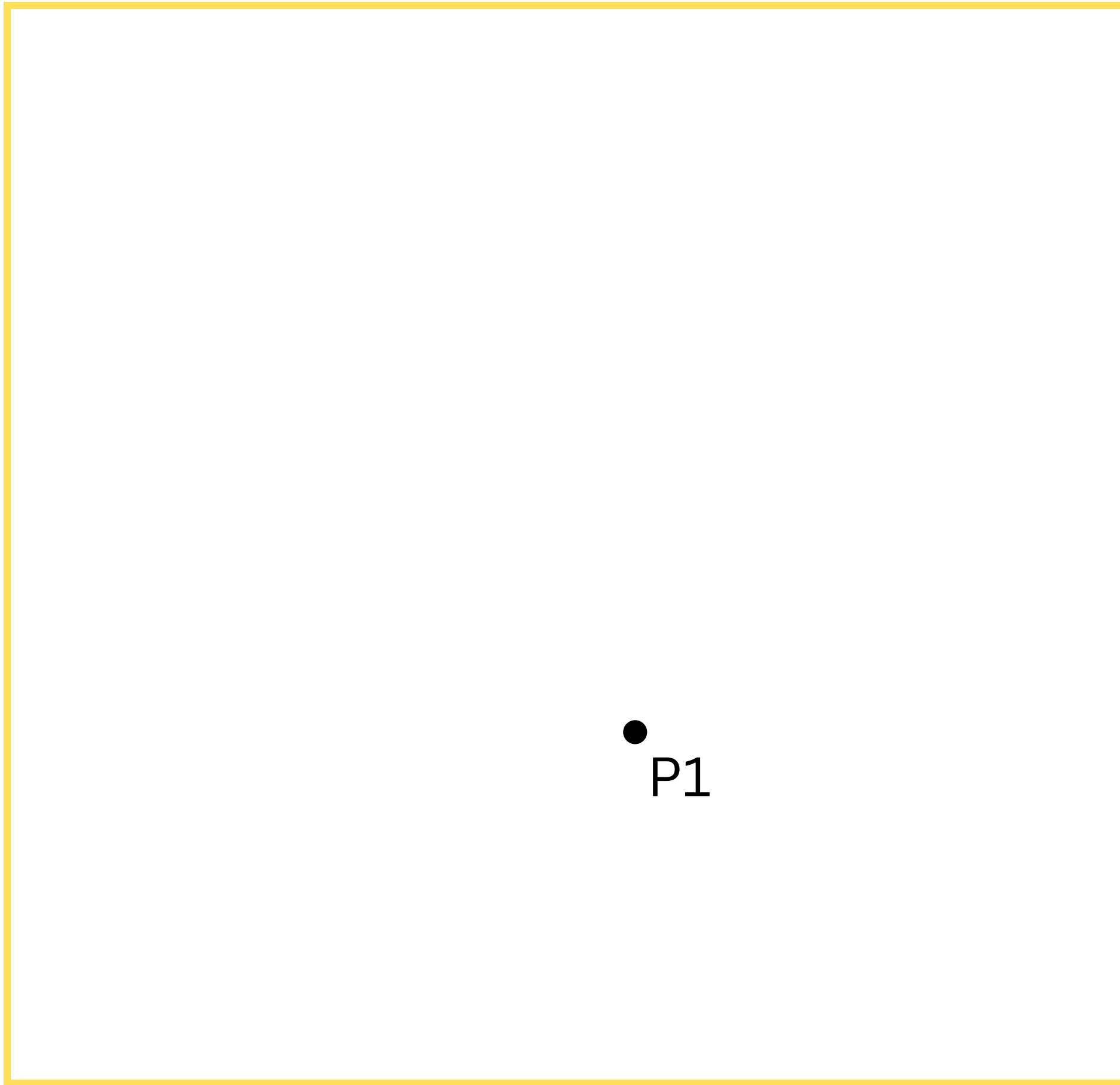
SE 

SW 

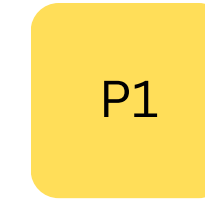
Capacity

IsDivided

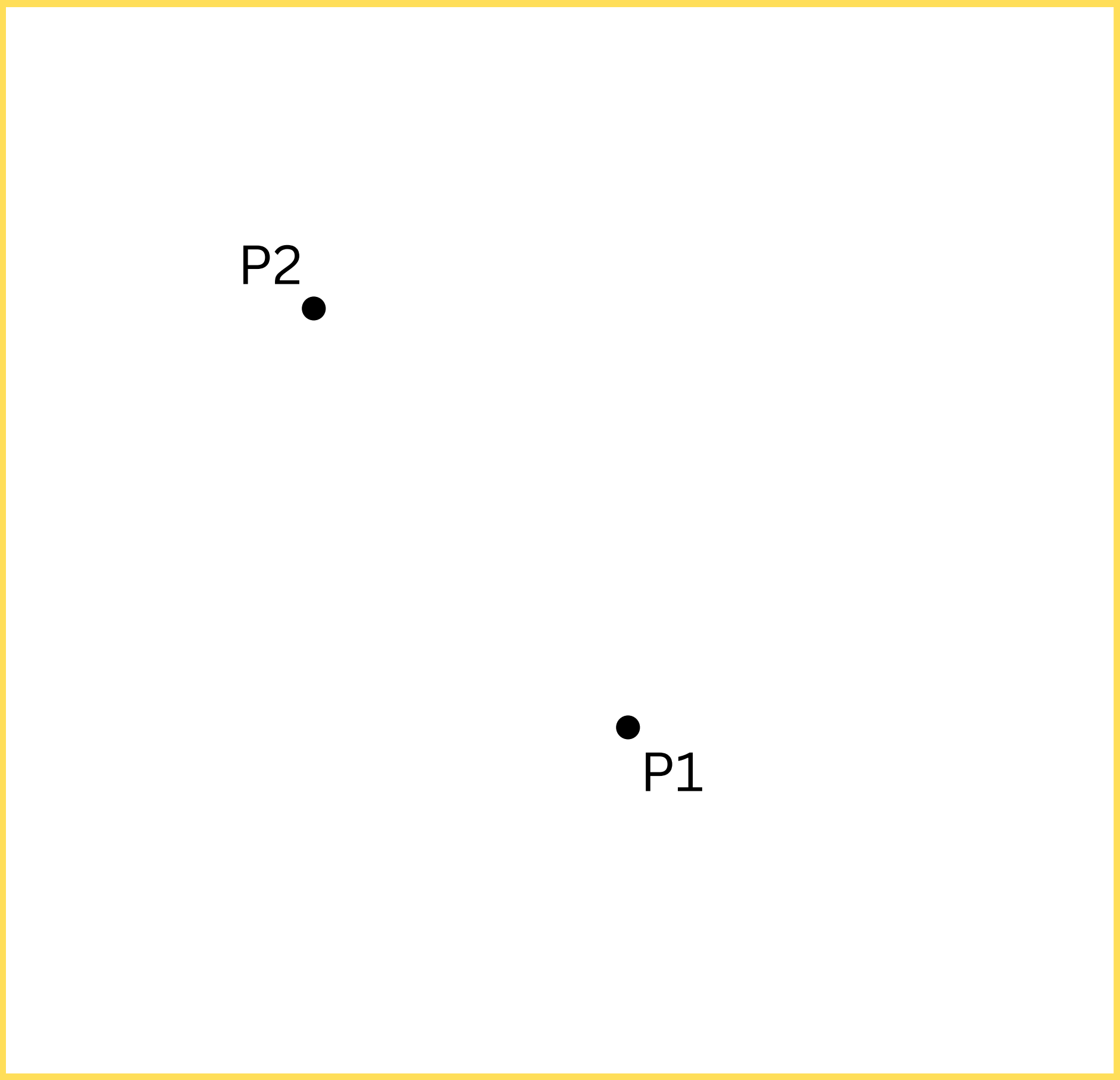
Inserting a Point in a QuadTree



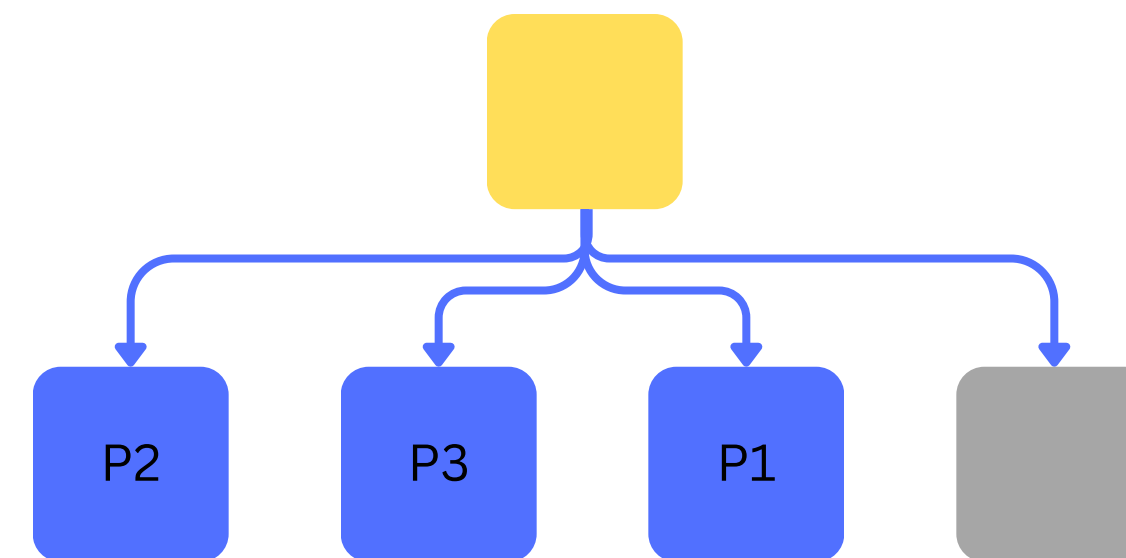
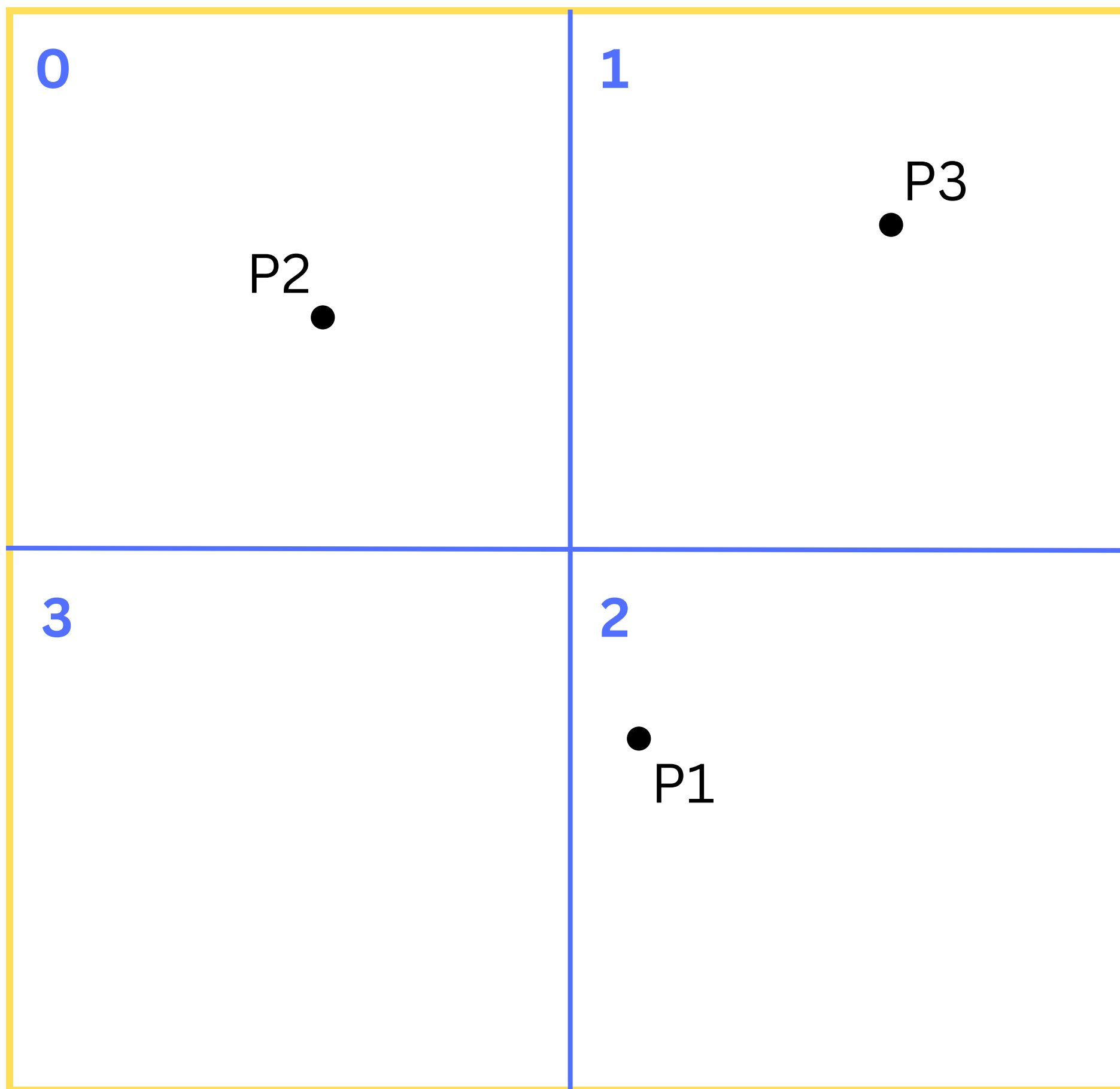
Capacity: 2

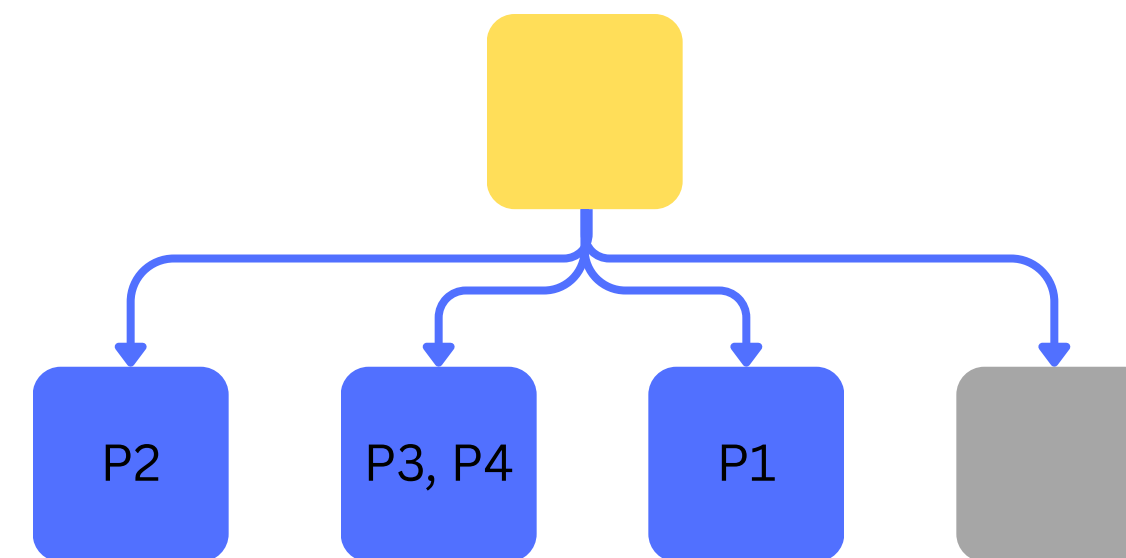
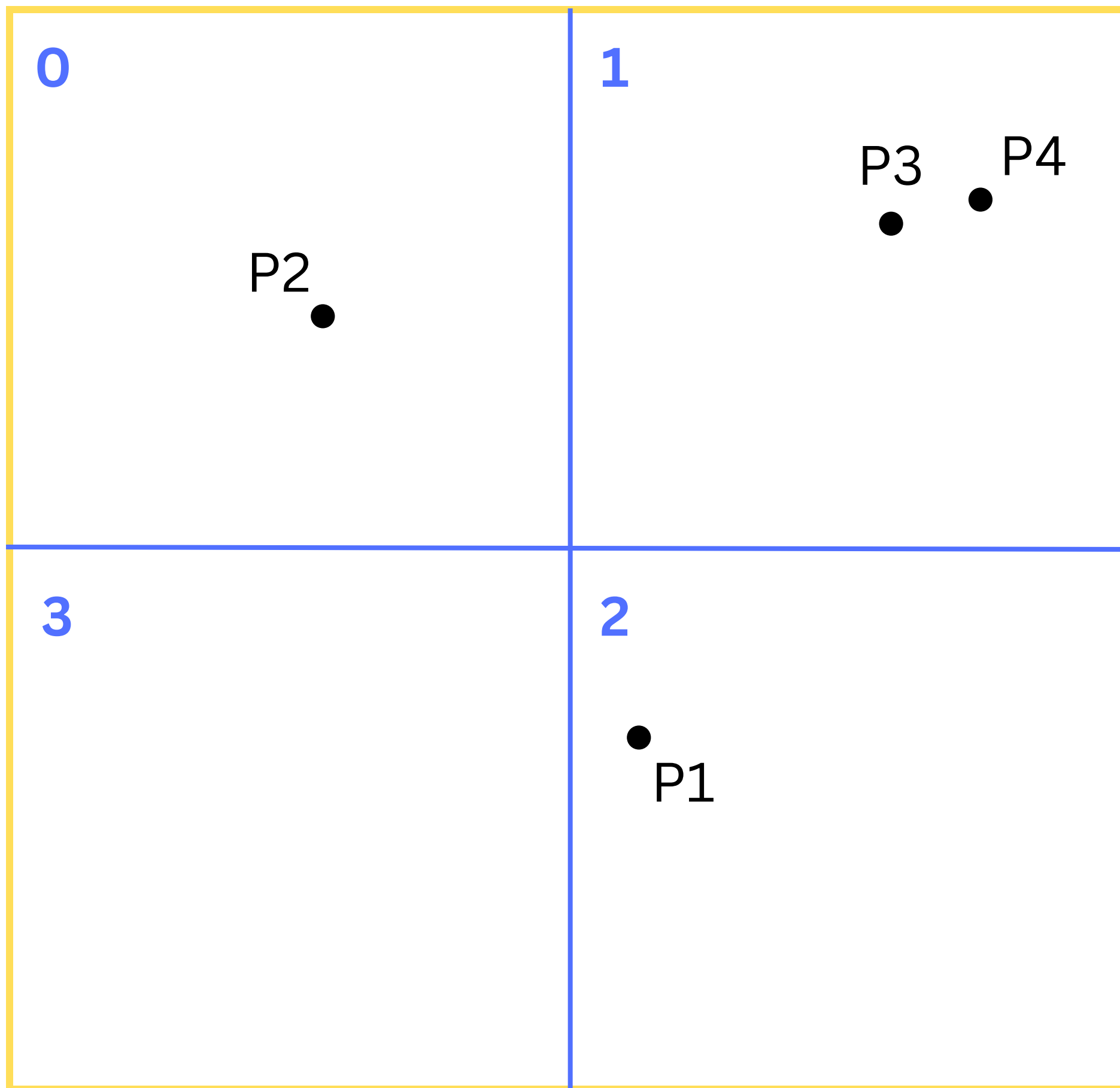


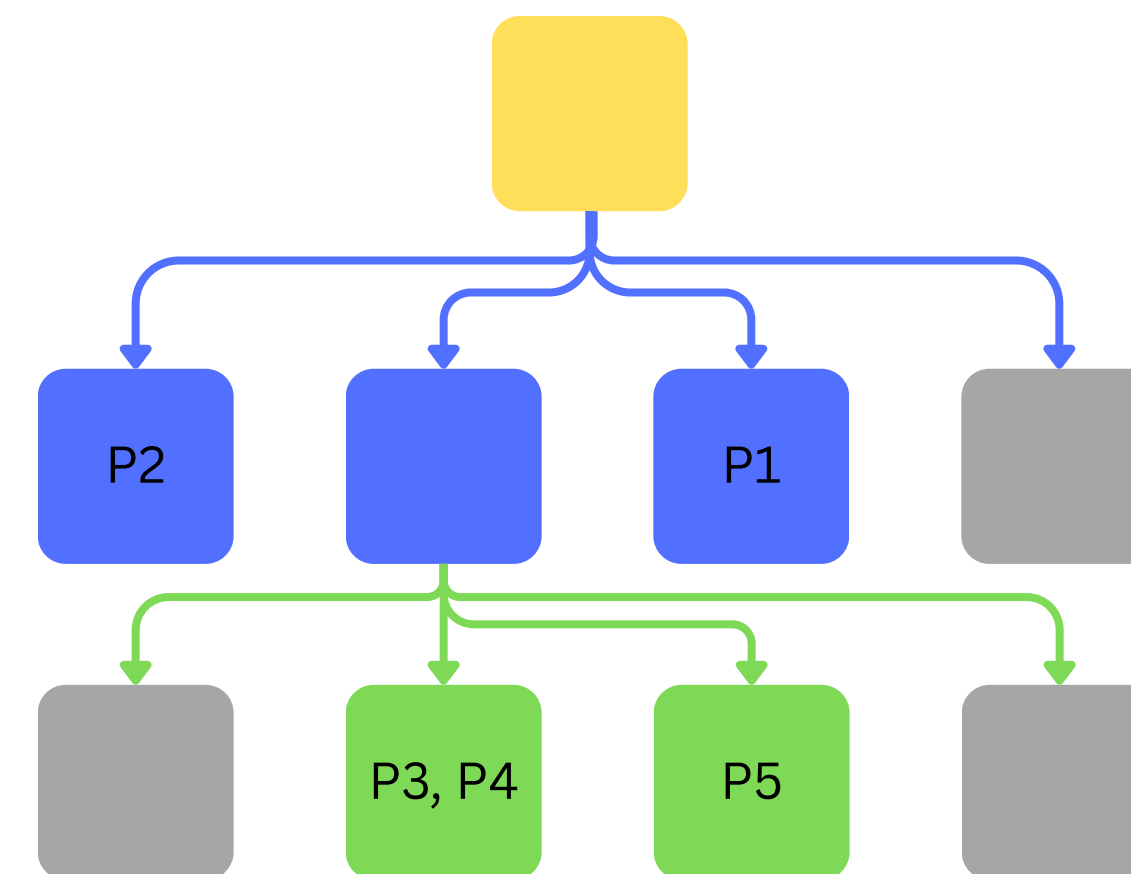
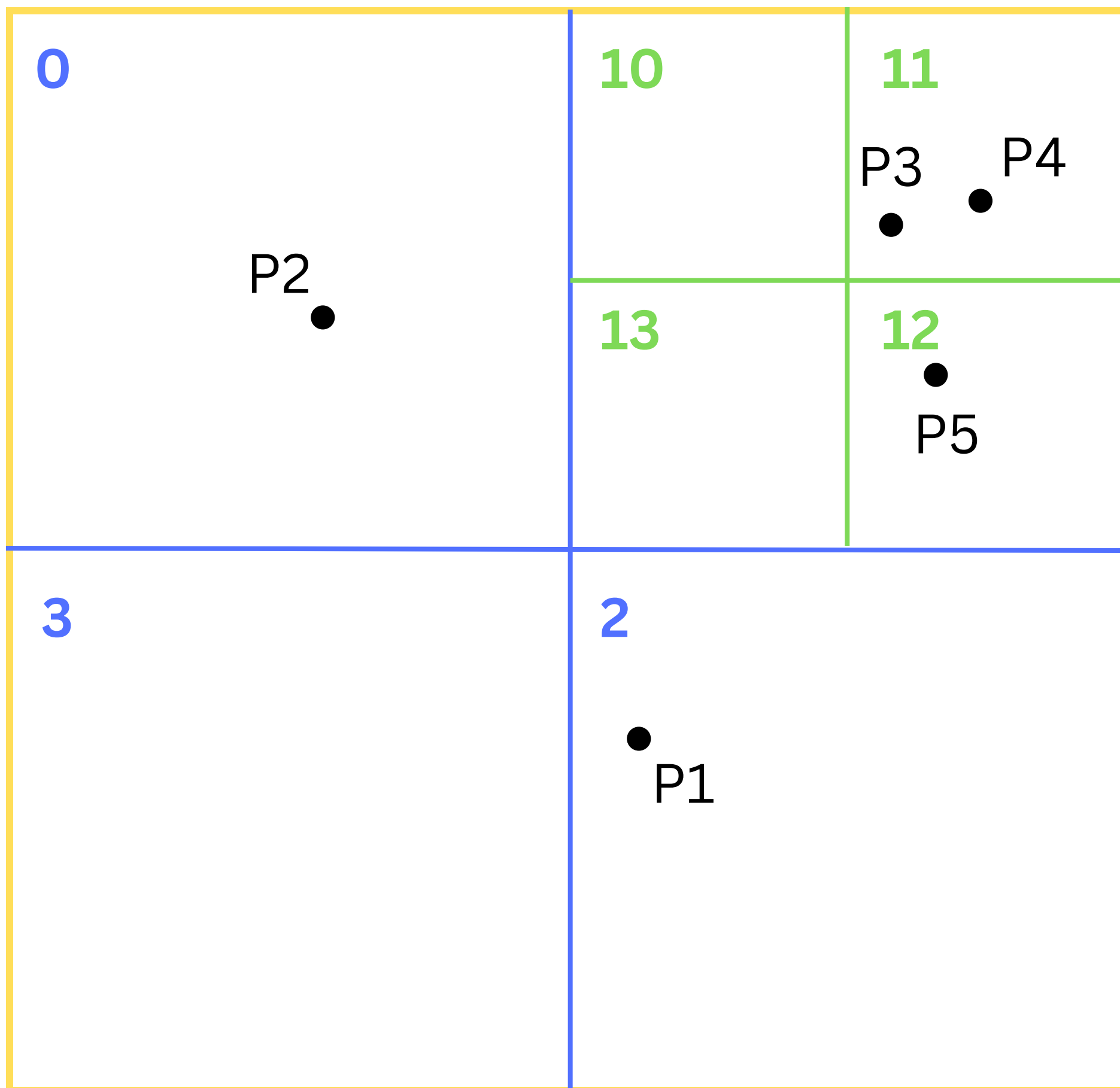
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P1



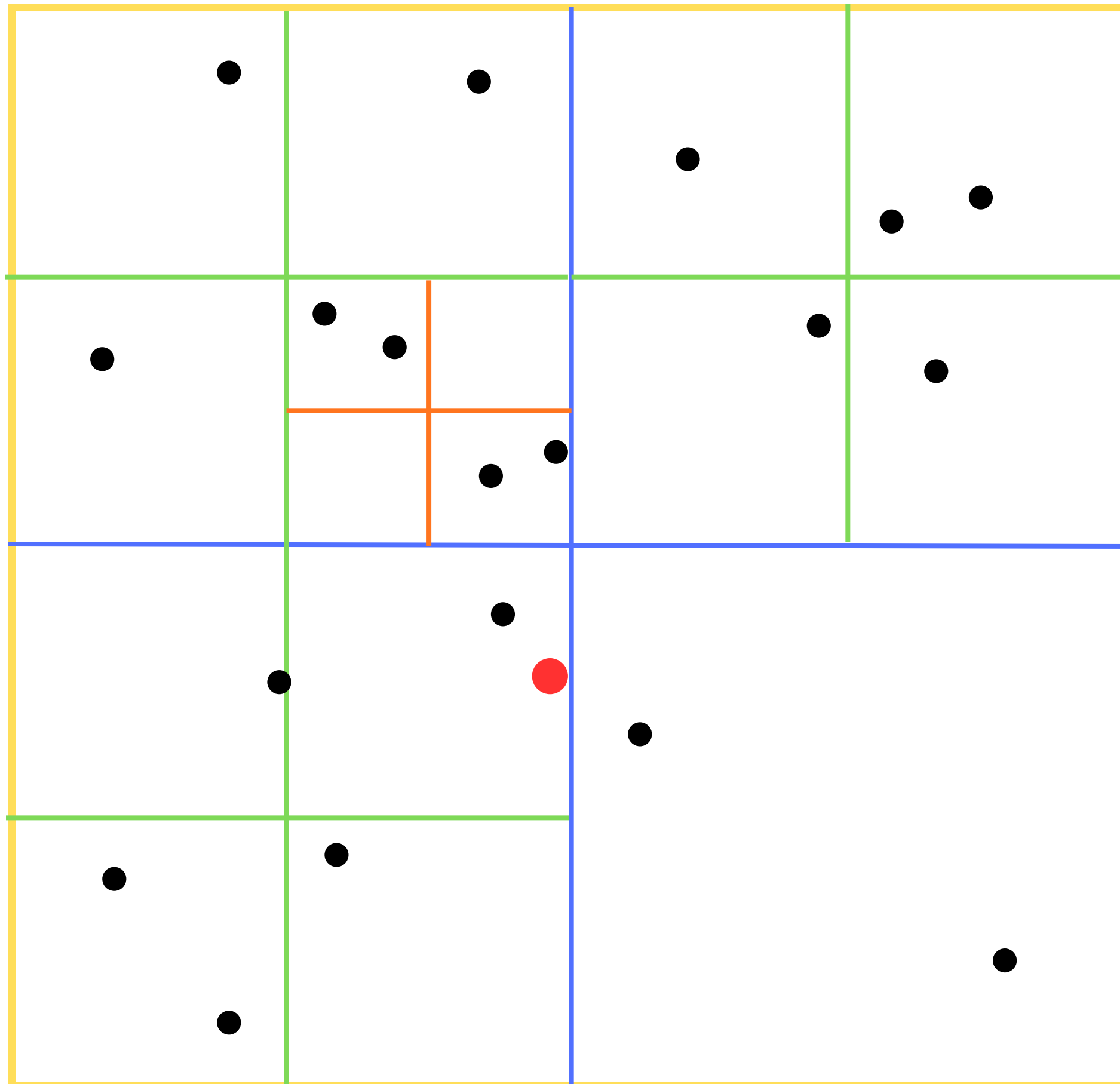
P1, P2



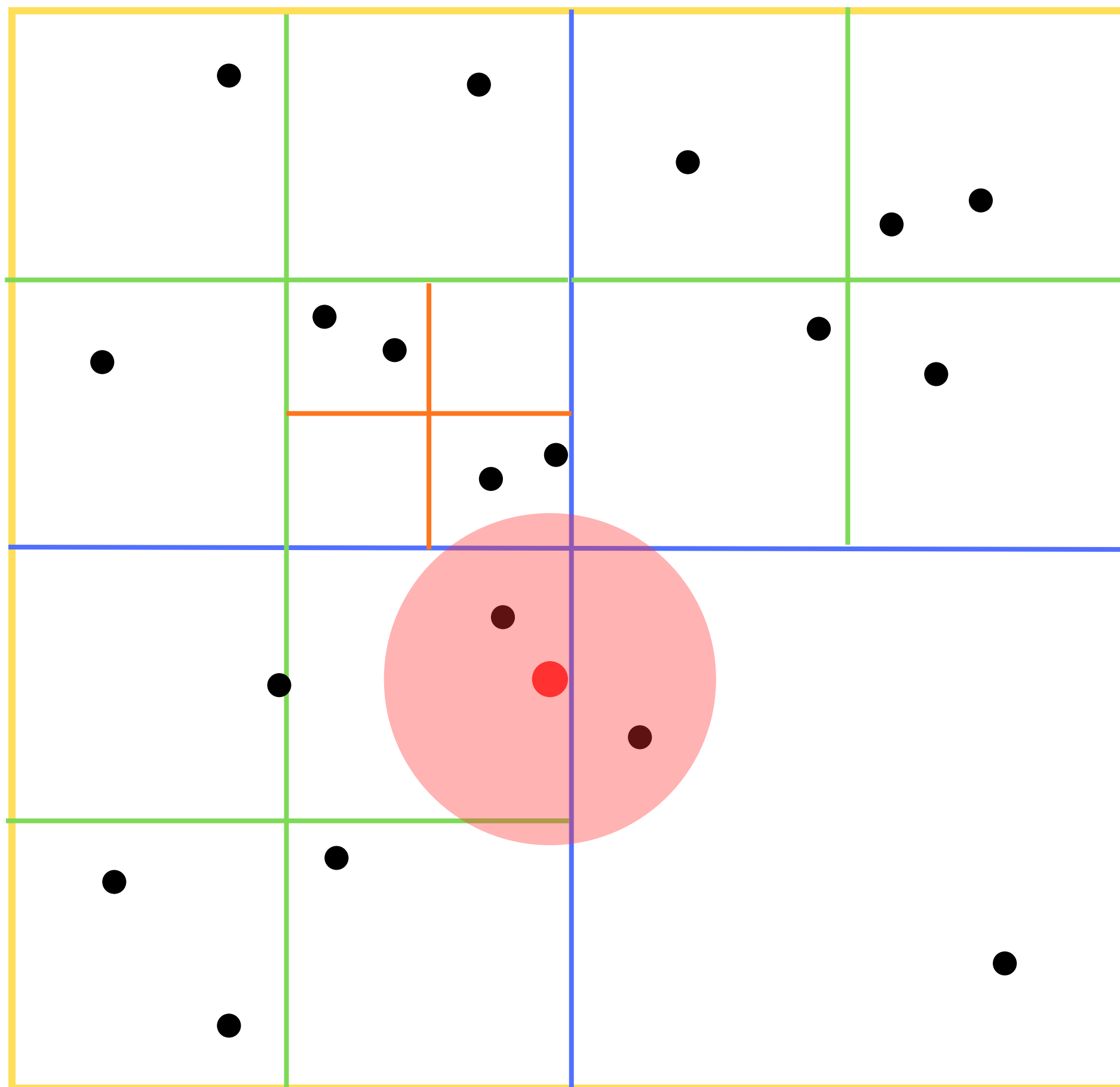




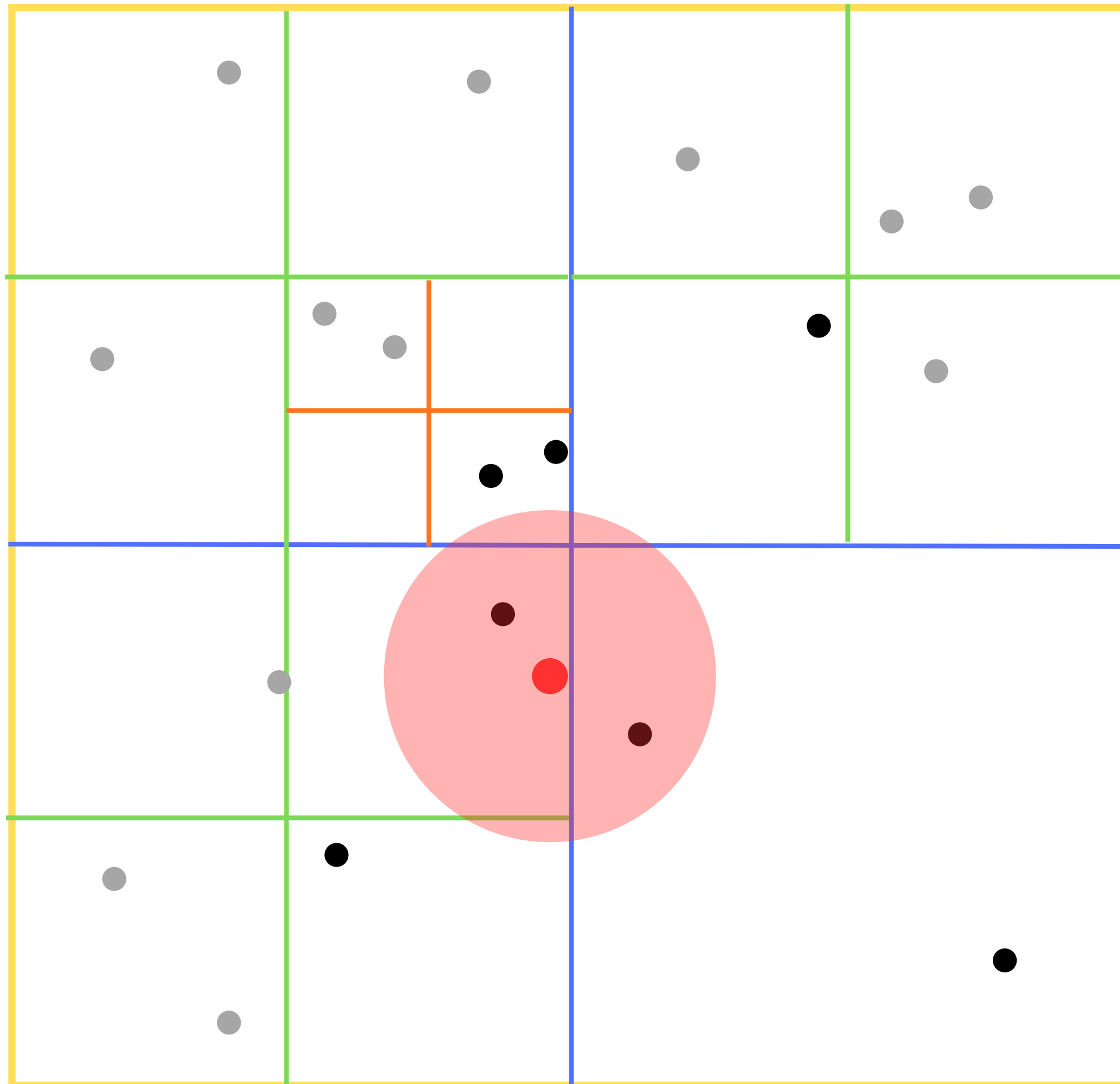
Range Query Algorithm



- Restaurant
- Your location



- Restaurant
- Your location
- Search area



- Restaurants searched
- Your location
- Search area
- Restaurants not searched

Use Cases

- 1. Game Development**
- 2. GIS and Mapping**

When NOT to use a QuadTree

- 1. Tree Degeneracy**
- 2. Dynamic Data**